

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: Study to Evaluate the Usability and Value of Integrated Digital Solutions in Medical Care of Participants With Multiple Sclerosis **Short Title/Acronym:** ILLUMINATE **Protocol Number:** MN44358 **NCT Number:** NCT05949580 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** MS Care Platform (icobrain ms and icompanion ms) **Phase of Development:** N/A (Non-interventional / Device usability study) **Medical Monitor:** Clinical Trials Director, icometrix / Hoffmann-La Roche

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the usability and value of an integrated multiple sclerosis (MS) care management platform in terms of improved monitoring of people with MS (pwMS) in routine clinical practice. **Secondary Objectives:** To assess patient medication adherence (MMAS-8), health literacy (HLQ), patient activation (PAM-13), level of platform engagement, and to quantify brain volume and abnormalities on MRI scans using the AI solution.

2.2 Background & Rationale Early initiation of effective disease-modifying therapies is crucial to prevent long-term disease progression in pwMS. Currently, diagnostic tools to assess clinical and subclinical disease activity may be used insufficiently. Therefore, there exists potential to implement technology-supported care paths designed to facilitate the detection of disease worsening and to improve interactions and consultations between patients and healthcare professionals (HCPs).

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Non-interventional) **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target **NS:** 500 patients **Number of Sites:** 10 sites across 5 countries **Estimated Study Start:** 01-Apr-2023 **Estimated Primary Completion:** 01-Apr-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with a definite diagnosis of Relapsing-Remitting Multiple Sclerosis (RRMS). 2. Time since MS diagnosis ≥ 1 year, and medical history recorded for at least 1 year before enrollment. 3. Expanded Disability Status Scale (EDSS) < 5.5 . 4. Able to comply with the study protocol, including having a compatible smartphone (Android version 5.0 or above, iOS version 11 or above), having a valid email address, and being willing to access the *icompanion ms* patient app regularly.

4.2 Exclusion Criteria 1. Inability or unwillingness to use a smartphone application or digital platform. 2. Lack of a compatible smartphone device preventing app installation and usage.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Daily/regular usage of the *icompanion ms* patient mobile app to track health and medication. Standardized quantitative analysis of brain MRI data at Baseline, 6, 12, 18, and 24 months via *icobrain ms*. **Route of Administration:** Digital Application (mHealth app) and cloud-based AI software (SaMD). **Duration of Treatment:** Up to 24 months (2 years of prospective data collection).

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care disease-modifying therapies for MS (e.g., Natalizumab, Ofatumumab, Alemtuzumab, Ocrelizumab). **Prohibited:** None specified.

6. Study Timeline & Visit Schedule

Visit ID	Window (Months)	Key Activities/Procedures
Screening/Baseline	Month 0	Informed Consent, Registration, App Download, Medical History, Baseline MRI, Baseline Questionnaires (SUS, HLQ, PAM-13)
Interim	Months 6, 12, 18	MRI scans (<i>icobrain ms</i> analysis), Patient Questionnaires (MMAS-8, PAM-13), HCP Portal Usage Tracking, Engagement Monitoring
End of Study	Month 24	Final MRI Analysis, Final Usability Questionnaires (SUS, MAUQ), Semi-structured Interviews, Exit

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Usability of the MS care management platform. **Measurement Method:** Assessed via the System Usability Scale (SUS) reported by HCPs (scores ranging from 0 to

100, with ≥ 50 indicating acceptable usability), and the mHealth App Usability Questionnaire (MAUQ) reported by patients.

7.2 Secondary & Exploratory Endpoints * Medication adherence measured by the 8-item Morisky Medication Adherence Scale (MMAS-8). * Patient health literacy and activation measured by the Health Literacy Questionnaire (HLQ) and Patient Activation Measure (PAM-13). * Number of MRI scans analyzed per center with icobrain ms and perceived changes in disease state evaluated by semi-structured interviews (SSIs).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine digital collection via the HCP portal and patient-reported changes in disease state. (Note: Due to the non-interventional, device-driven nature of the study, physical AEs related to the intervention itself are not anticipated). **Serious Adverse Events (SAEs):** Reported within 24 hours of site/HCP awareness in accordance with routine pharmacovigilance for patients' background MS therapies. **Data Monitoring Committee (DMC):** Overseen by Hoffmann-La Roche and icometrix.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population encompassing all patients enrolled who have successfully registered on the *icompanion ms* app. **Sample Size Justification:** $N=500$ was selected to provide robust descriptive statistics and qualitative feedback on platform usability and value across international medical centers. **Handling of Missing Data:** Missing questionnaire items will be handled per the scoring guidelines of the specific scales (e.g., SUS, MMAS-8).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP), the Declaration of Helsinki, and applicable data privacy regulations (e.g., GDPR) for cloud-based medical data. **Monitoring Plan:** Continuous backend monitoring of app engagement (assessed monthly by the number of logins and time spent) and HCP portal utilization. **Audit Trail:** Secure, cloud-based data logs tracking platform activity, MRI uploads to icobrain ms, and electronic patient-reported outcomes (ePROs).

11. Study Identifiers & Governance

Primary ID: MN44358 **Registry Number:** NCT05949580 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** ILLUMINATE **Official Regulatory Title:** Study to Evaluate the Usability and Value of Integrated Digital Solutions in Medical Care of Participants With Multiple Sclerosis

12. Clinical Framework (MS Specific)

Primary Condition: Relapsing-Remitting Multiple Sclerosis (RRMS) **Diagnosis Threshold:** Definite RRMS diagnosis ≥ 1 year, EDSS < 5.5 **Medical Methodology:** Integrated Digital Solutions / mHealth & SaMD **Optimization Target:** Improved monitoring of disease progression and patient-HCP interaction

13. Study Procedures & Methodology

Management Pathway: Technology-supported MS care paths **Education Protocol (HCP):** Portal onboarding and digital tool integration training **Education Protocol (Patient):** App download, registration, and usage instructions **Quality Audit Mechanism:** Engagement monitoring (logins/time spent) & Usability Questionnaires (SUS/MAUQ)

14. Observation & Follow-Up Schedule

Total Duration: 104 Follow-up weeks (24 months) **Onsite Visit Cadence:** Months 0, 6, 12, 18, 24 for MRI/clinical consultations **Remote Monitoring Frequency:** Continuous via mobile app, evaluated monthly **Checklist Review Interval:** Every 6 months (Months 6, 12, 18, 24)

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				